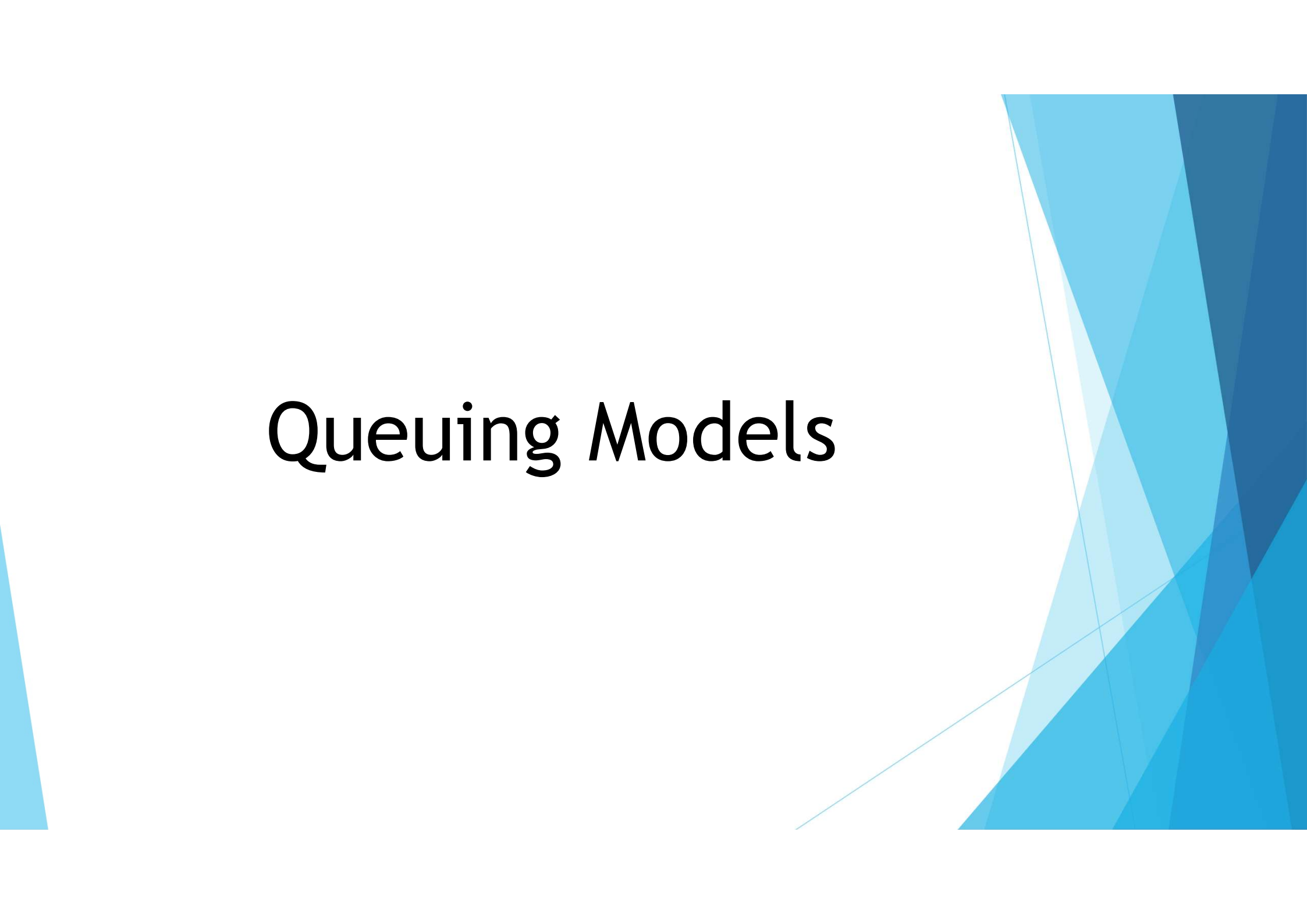


OPERATIONAL RESEARCH



Queuing Models



Definitions

- ▶ **Queuing Theory**
- ▶ Queuing Theory is a collection of Mathematical models of various queuing systems.
- ▶ It is based on probability concepts. It gives indication of the **capability** of given system and the possible changes in its performance with modification to the system.
- ▶ The application of the queuing theory cannot be viewed as optimization process because it does not Maximization or Minimization Objective Function.
- ▶ **Queues** or **Waiting Lines** arise when the demand for a service facility exceeds the capacity of that facility.

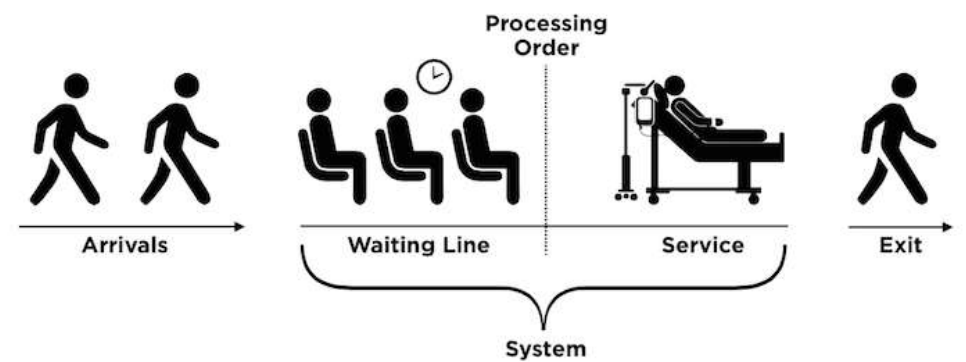
Need to learn Queuing Theory

- ▶ To find out the cost of offering the service
- ▶ To find out cost incurred due to DELAY in service
- ▶ To allocate the resources

Operating Characteristics of Queuing System

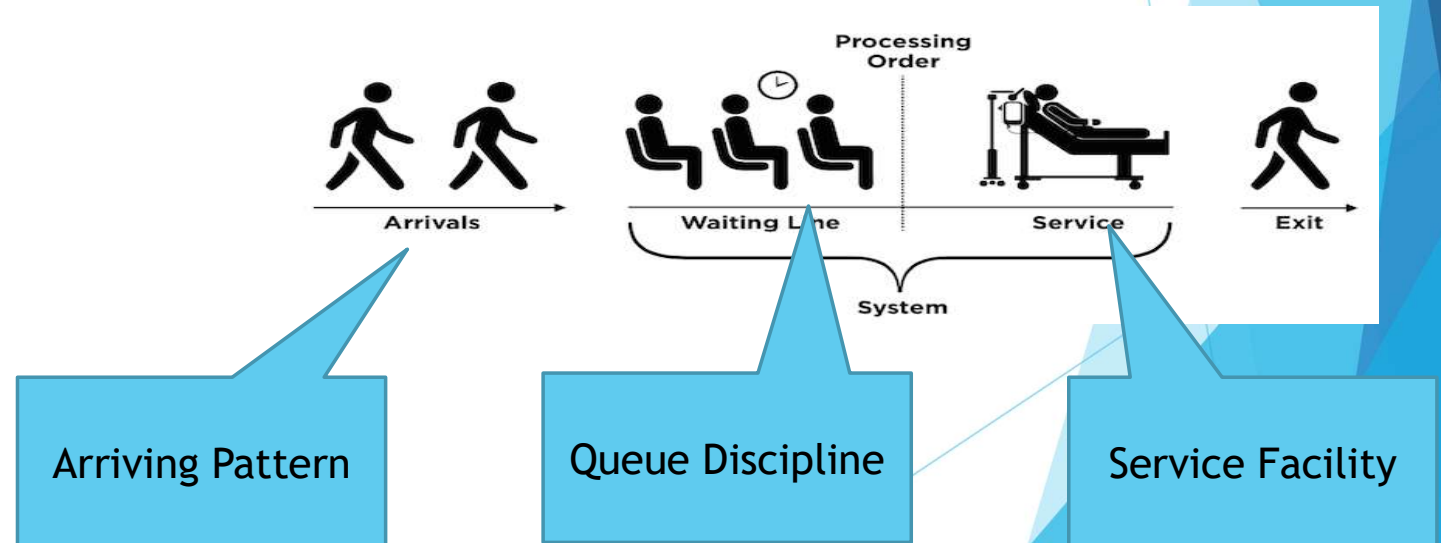
Queuing model has following characteristics which help to understand and manage a queue

- ▶ **Queue Length :** The number of customers in the awaiting line reflects one of the two conditions. Short Queues & Long Queues.
- ▶ **Number of customers in the System :** The number of customers in queue and also those being served in the queue relates to the service efficiency and capacity. Large value implies congestion, potential customer dissatisfaction and need for more capacity.
- ▶ **Waiting time in Queue :** Time to wait for service. Waiting time seems long to customers, they perceive that the quality of service is poor, Long waiting times may indicate a need to adjust the service rate of the system or change the arrival rate of customers.
- ▶ **Waiting time in system :** The total elapsed time from entry into the system until the exit from the system. This will indicate problems with customers, server efficiency or capacity.
- ▶ **Service Facility Utilization :** The collective utilization of the service facilities reflects the percentage of time the facilities are busy.



Constituents/Components of Queuing System

- ▶ **Arriving Pattern** ▶ It is the average rate at which customers arrive.
- ▶ **Queue Discipline** ▶ It is the common method of choosing the customer for service amongst those waiting for service is “**First Come First Serve**”
- ▶ **Service Facility** ▶ Examining the number of customers served at a time and the statistical pattern of time taken for service at the service facility.



Constituents/Components of Queuing System

▶ Arriving Pattern

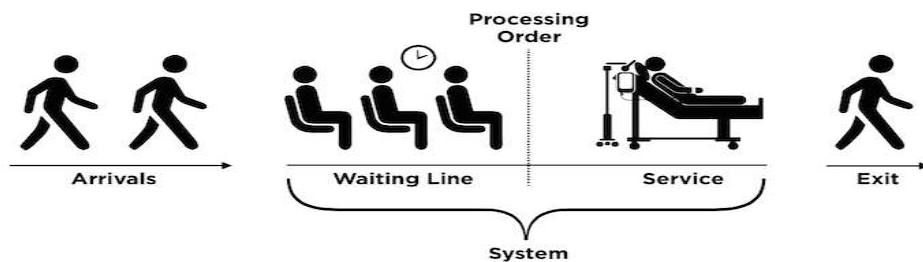
- ▶ Size of population / Calling Population : Unlimited (Infinite) or Limited (Finite)
- ▶ Pattern of Arrivals
 - ▶ When the number of arrivals of customers is completely random, then they follow a **Poisson Distribution**. Which is average number of arrivals per unit time.
- ▶ Behavior of Arrivals

Constituents/Components of Queuing System

► Queue Discipline - Static Queuing

- Static Queuing Discipline : Based on Individual customer's status

First In First Out



Last In First Out



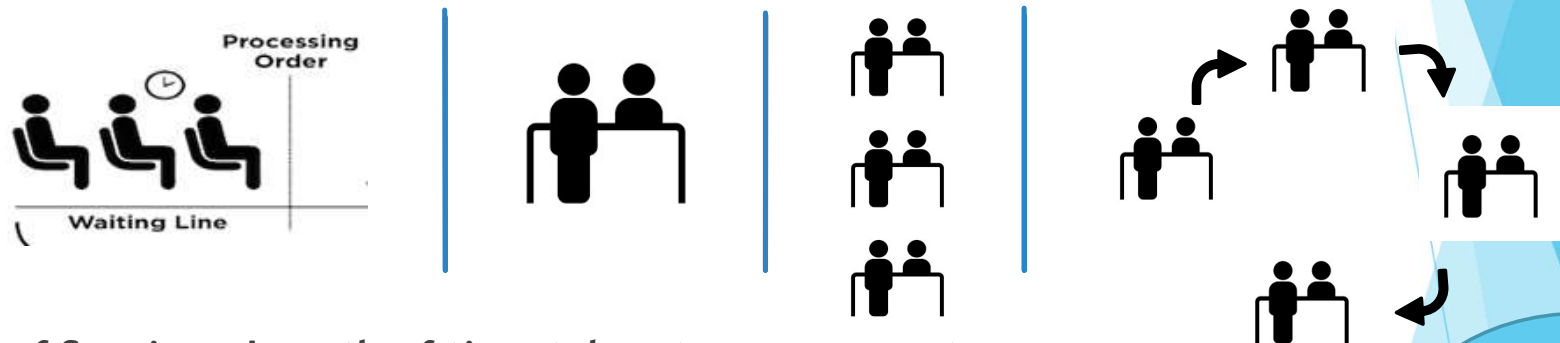
► Queue Discipline - Dynamic Queuing

- **Service in Random Order (SIRO)** : Customers are selected at random irrespective of their arrival time
- **Priority Service** : Customers are grouped in priority classes on basis of some attributes such as service time or urgency, a FIFO is used for each class for servicing.
- **Pre-emptive priority (or Emergency)** : Customers are served immediately on priority after arrival, while the low priority customer still waiting in Queue

Constituents/Components of Queuing System

► Service Facility

- Availability of service : Understand/Analyze the constraints to reduce the number of customers
- Number of Service Centers
 - Number of service centers are defined based on the service offered



- Duration of Service : Length of time taken to serve a customer.

- Constant Service Time : Not in practice
- Completely Random Service Time :
 - Server does not distinguish between various arrivals
 - Server does not change the service time based on previous arrival
 - Server forgets the time for which he/she serving the customer

Service Time follows Erlang Distribution

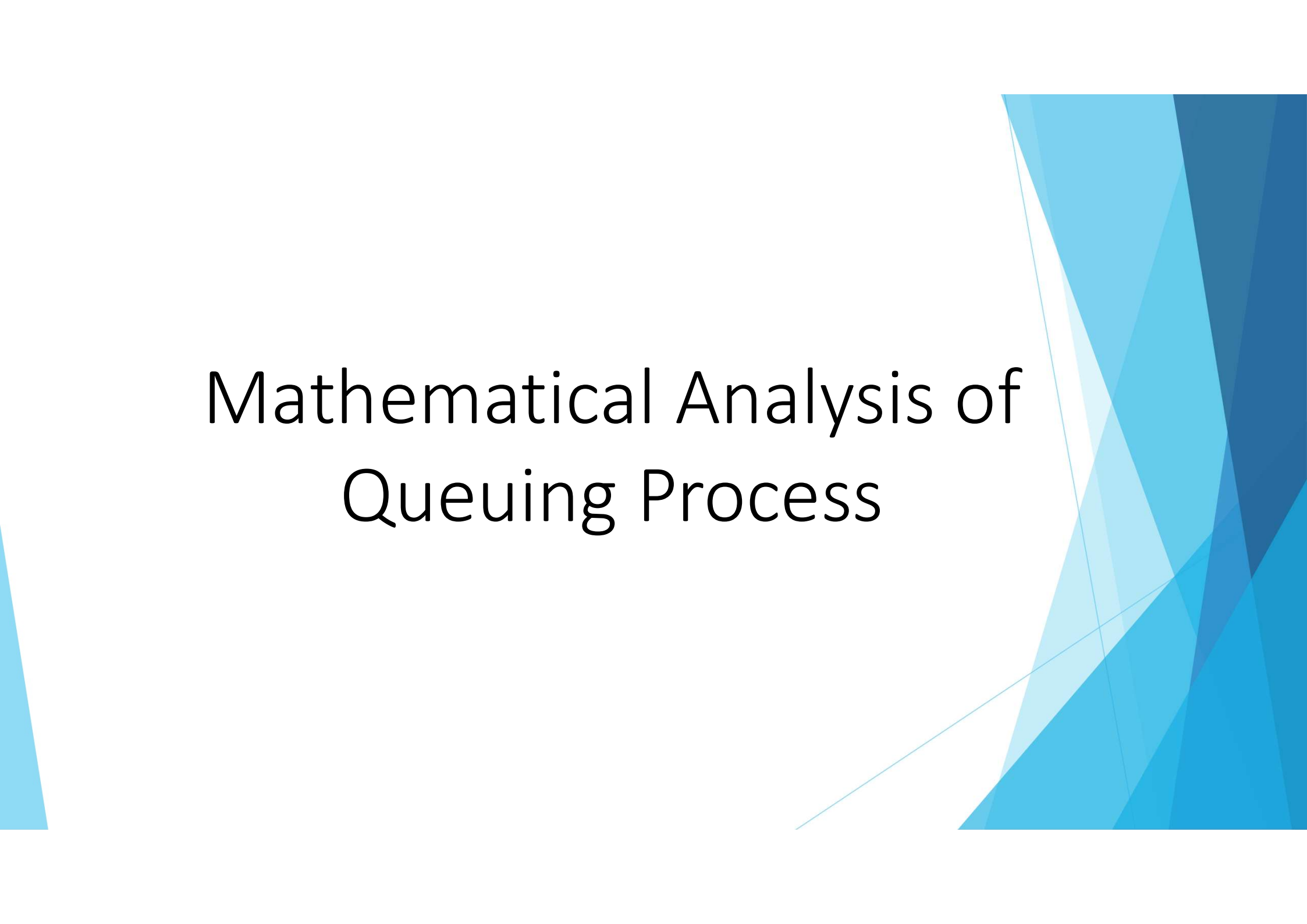
Exponential Distribution

Constituents/Components of Queuing System

▶ Customer Behavior

- ▶ **Balking:** Arriving customer are said to “**Balk**” if they do not join the queue because of their relevance to wait or some customers even before joining the queue get discouraged by seeing the number of customers already in the service system or estimating the excessive waiting time for a desired service, decide to return for service at later time. In queuing theory it is known as “**Balking**”.
- ▶ **Collusion:** Customers may be in collusion, meaning that only one person will join the queue but will demand service on behalf of several customers.
- ▶ **Reneging:** Impatient customers who would not wait beyond a certain time and leave the queue are said to renege.
- ▶ **Jockeying:** Some customers keep on switching over from one queue to another in a multiple service centers. This is called “**Jockeying**”.
- ▶ **Pattern of Arrival :**
 - ▶ Customers can arrive individually or in batches.
 - ▶ May arrive in scheduled or unscheduled time
 - ▶ **Poisson Distribution :** Probabilities for the number of customers that may arrive in specific interval of time λ
 - ▶ **Exponential Distribution :** Probabilities for times gap between two consecutive arrivals μ

Mathematical Analysis of Queuing Process



Properties of Queuing System

- ▶ **Statistical Equilibrium**: The probability distribution of the arrivals will then be different than of the initial state. The State in which the probability distribution remains the same, is called Steady State and the system is said to have acquired a state of **Statistical Equilibrium**.

- ▶ **Properties of Queuing System**



- ▶ **Arrivals**

- ▶ Customers are discrete entities
- ▶ Population – Finite/Infinite
- ▶ No Simultaneous arrivals
- ▶ Pattern of arrivals

- ▶ **Queuing Discipline**

- ▶ First Come First Serve



- ▶ **Service**

- ▶ Single Serve Channel / Multiple Channels
- ▶ Single Queue / Infinite capacity

Properties of Queuing System - Service

► Two Aspects of Service System

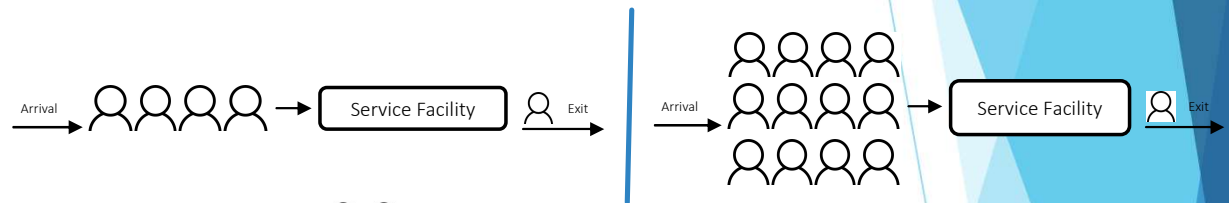
- Configuration of the service System
- Speed of Service

► Configuration of Service System

- Configuration of service system means how the service facilities exists
- Mostly depends on number of channels or number of servers

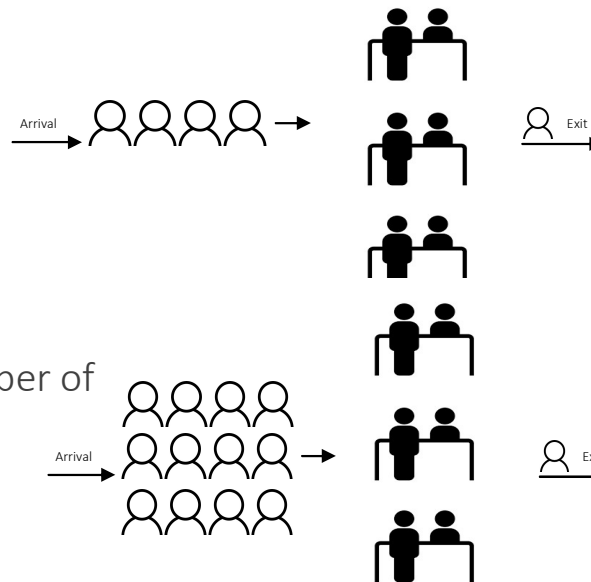
► Single Server Single Queue

- One Queue One Service Station



► Several Servers Single Queue

- More than one server, provides same type of service



► Service facilities in Series

- Portion of service at each server

► Several Servers Several Queues

- More than one server, more number of queues, can be different service



Properties of Queuing System – Speed of Service

- ▶ Expressed as
 - ▶ Service Rate
 - ▶ Service Time
- ▶ Service Rate
 - ▶ It is defined as the number of customers serviced during a particular time period
- ▶ Service Time
 - ▶ It is defined as the amount of time needed to serve a single customer
- ▶ Service Time & Service Rate are reciprocal to each other and either of them is sufficient to indicate the capacity of the facility
- ▶ For e.g. If a cashier can attend an customer on an average of 5 customers in an hour, what will be the Service Rate & Service Time ?
- ▶ Service Rate : 5 customers / hour,
- ▶ Service Time : $60/5 = 12$ Minutes / customer

Arrival of Customers

- ▶ **Average / Expected or Mean Arrival Rate** λ
 - ▶ **Arrival Rate** : The rate at which customers are coming to get the services
 - ▶ **Average Arrival Rate** : The number of customers **arrive** to avail services in a given amount of time
- ▶ **Calculate Average Arrival Rate**
 - ▶ Customers are arriving at every 6 minutes. Calculate the average arrival rate for 1 hour.

▶ **Average Arrival Rate**
 $\lambda = 10$

Serving of customers

- ▶ **Average / Expected or Mean Service Rate** μ
 - ▶ **Service Rate** : The rate at which customers are serviced by the service provider
 - ▶ **Average Service Rate** : The number of customers **served** in a given amount of time
- ▶ **Calculate Average Service Rate**
 - ▶ On an average, the server takes 5 minutes to serve their customers. Calculate the average service rate for 1 hour.

▶ **Average Service Rate**
 $\mu = 12$

Utilization Rate

$$\mathbf{R} = \frac{\text{Average Arrival Rate}}{\text{Average Service Rate}} = \frac{\lambda}{\mu} = \frac{10}{12} = 0.833$$

► Scenarios for Utilization Rate

$$\lambda > \mu$$

- Infinite Queue
- Busy Service Provider
- Failure of Service system

$$\lambda = \mu$$

- No Queue Formation
- Server will be busy
- Consumer comes and gets served

$$\lambda < \mu$$

- No Queue Formation
- Idle time for server
- Customer gets served as soon as in Queue

Probability of exactly “n” Customers in System

- ▶ Probability of No Customer ($n = 0$) – Idle Rate

$$P_{(0)} = 1 - R = \left(1 - \frac{\lambda}{\mu}\right)$$

- ▶ Probability of One Customer ($n = 1$)

$$P_{(1)} = R * P_{(0)} = R * (1 - R) = \frac{\lambda}{\mu} * \left(1 - \frac{\lambda}{\mu}\right)$$

- ▶ Probability of Two Customers ($n = 2$)

$$P_{(2)} = R * P_{(1)} = R * R * P_{(0)} = R * R * (1 - R) = \frac{\lambda}{\mu} * \frac{\lambda}{\mu} * \left(1 - \frac{\lambda}{\mu}\right)$$

- ▶ Probability of “n” Customers

$$P_{(n)} = R^n * P_{(0)} = \left(\frac{\lambda}{\mu}\right)^n * \left(1 - \frac{\lambda}{\mu}\right)$$

- Average Number of customers in System L_s (Length of System)

$$L_s = \frac{\text{Utilization Rate}}{\text{Idle Rate}} = \frac{\lambda}{\mu} / \left(1 - \frac{\lambda}{\mu}\right) = \frac{\lambda}{\mu - \lambda}$$

- Average Number of Customer in Queue L_q (Length of Queue)

$$L_q = L_s - \text{Utilization Rate} = \frac{\lambda}{\mu - \lambda} - \frac{\lambda}{\mu} = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

- Average waiting time for customer in Queue W_q

$$W_q = \frac{\text{Avg Number of customers in the Queue}}{\text{Avg Arrival Rate}} = \frac{L_q}{\lambda} = \frac{\lambda^2}{\mu(\mu - \lambda)} / \lambda = \frac{\lambda}{\mu(\mu - \lambda)}$$

- Average waiting time for customer in System W_s

$$W_s = \frac{\text{Avg Number of customers in the System}}{\text{Avg Arrival Rate}} = \frac{L_s}{\lambda} = \frac{\lambda}{\mu - \lambda} / \lambda = \frac{1}{\mu - \lambda}$$

► Expected Length of non-Empty Queue

$$L_{q'} = \frac{\text{Expected Number of customers in Queue}}{P(\text{More than 1 customer in system})} = \frac{L_q}{P(n > 1)} = \frac{\lambda^2}{\mu(\mu - \lambda)} \bigg/ \left(\frac{\lambda}{\mu} \right)^2 = \frac{\mu}{\mu - \lambda}$$

Models of Queuing Theory

► Single Server Model

► Multi Server Model

$$\begin{array}{l|l|l} L_s = \frac{\lambda}{\mu - \lambda} & L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} & L_{q'} = \frac{\mu}{\mu - \lambda} \\ W_s = \frac{1}{\mu - \lambda} & W_q = \frac{\lambda}{\mu(\mu - \lambda)} & \end{array}$$

Example 1

On the desk of an office of a Banking Company, the arrivals of the customers follow a poisson distribution and on an average at every 10 minutes a customer arrives. The officer responsible takes an average 6 minutes to serve a customer, assuming the exponentially distributed.

Find out the average arrival rate for λ

- a. 1 hour
- b. 15 minutes
- c. 8 hours

Find out the average Service rate for μ

- a. 1 hour
- b. 15 minutes
- c. 8 hours

Find out the average arrival rate for

- a. 1 hour $\lambda = 6$
- b. 15 minutes $\lambda = 1.5$
- c. 8 hours $\lambda = 48$

Find out the average Service rate for

- a. 1 hour $\mu = 10$
- b. 15 minutes $\mu = 2.5$
- c. 8 hours $\mu = 80$

Example 1

Find out the Utilization rate for

- a. 1 hour
- b. 15 minutes
- c. 8 hours

$$R = \frac{\lambda}{\mu}$$

Find out the average arrival rate for

- a. 1 hour **R = 0.6**
- b. 15 minutes **R = 0.6**
- c. 8 hours **R = 0.6**

Find out the

- a. Idle Rate
- b. Prob (System is Idle)
- c. Average Idle time of Server

$$P_{(0)} = 1 - R = 0.4$$

$$P_{(0)} = 0.4$$

$$P_{(0)} * t = 0.4 * 8 \text{ hr} = 3.2 \text{ hr}$$

Example 1

Find out the Prob (exactly "n" customers in System)

a. 1 hour & $n = 5$

$$P^{(n)} = \left(\frac{\lambda}{\mu} \right)^n * \left(1 - \frac{\lambda}{\mu} \right)$$

Answer

$$P^{(5)} = \left(\frac{6}{10} \right)^5 * \left(1 - \frac{6}{10} \right)$$

$$P^{(5)} = 0.0311$$

Find out the Number of customers in Sysem (Length of System)

a. 8 hours

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{48}{80 - 48} = \frac{3}{2}$$

Find out the Number of customers in Queue (Length of Queue)

a. 8 hours

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)} = 0.9$$

Example 1

Find out the Waiting time in System

a. 8 hour

$$W_s = \frac{1}{\mu - \lambda} = 0.0312$$

Find out the Expected Waiting Time in Queue

a. 8 hours

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = 0.0187$$

Find out the Expected length of non-Empty Queue

a. 8 hours

$$L_{q'} = \frac{\mu}{\mu - \lambda} = 2.5$$

Example 2

Arrivals of customers to payment counter in a bank follow poisson distribution with an average of 10 per hour. The service time follows negative Exponential Distribution with an average of 4 minutes.

What is the average number of customers in Queue ?

The bank will open one more counter when the waiting time of a customer is at least 10 minutes. By how much the flow of arrivals should increase in order to justify second counter ?

$$\lambda = 10 \text{ per hour}$$

$$\mu = 4 \text{ per minutes} = \text{how much per hour ?} = 15 \text{ per hour}$$

What is the average number of customers in Queue ?

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

$$L_q = \frac{4}{3}$$

Example 2

The bank will open one more counter when the waiting time of a customer is at least 10 minutes. By how much the flow of arrivals should increase in order to justify second counter ?

waiting time of a customer is at least 10 minutes.

$$W_q = \frac{\lambda}{\mu(\mu - \lambda)} = 10 \text{ minutes}$$

Convert it into Hour

$$= \frac{1}{6}$$

By how much the flow of arrivals should increase ? λ

$$W_q = \frac{\lambda}{15(15 - \lambda)} \quad \lambda = 10.7$$

By how much the flow of arrivals should increase ?

Previous Arrival Rate = 10
New Arrival Rate = 10.7
Increase by 0.7

Example 3

(B) A self service store employs one cashier at its counter. 8 customers arrive on an average every 5 minutes while the cashier can serve 10 customers in the same time. Assuming Poisson distribution for arrival and exponential distribution for service rate, determine:

- i) Average number of customer in the system.
- ii) Average number of customer in queue.
- iii) Probability that there is no customer at the counter
- iv) Probability that there are equal to or more than 3 customers in the queue.
- v) Probability that there are more than 2 customers in the system
- vi) Probability that an arrival finds that 2 persons are waiting for there turn.